

5

ALGEBRA

Figure 5.1 In these algebraic equations, the x represents different numbers. (credit: Larissa Chu, CC BY 4.0)

Chapter Outline

- 5.1 Algebraic Expressions
- 5.2 Linear Equations in One Variable with Applications
- 5.3 Linear Inequalities in One Variable with Applications
- 5.4 Ratios and Proportions
- 5.5 Graphing Linear Equations and Inequalities
- 5.6 Quadratic Equations In One Variable with Applications
- 5.7 Functions
- 5.8 Graphing Functions
- 5.9 Systems of Linear Equations in Two Variables
- 5.10 Systems of Linear Inequalities in Two Variables
- 5.11 Linear Programming



Introduction

The jump from arithmetic to algebra can be a difficult one for many students. Many students struggle with the idea that mathematics can include situations that aren't static and do change. In elementary arithmetic, a situation such as:

$$5 + 3 = \underline{\quad}$$

is a static situation and will yield the answer of 8 every time. However, a situation such as: $5x + 3 = \underline{\quad}$

can yield many different answers because the answer depends on what amount (number) that x represents. Since the value of x can vary (represent different values), it is known as a variable.

Algebra is useful to better model real life situations. In the first equation shown, $5 + 3 = \underline{\quad}$ can only model situations where you add those two numbers together. For example, if your uncle gives you five dollars and your aunt gives you three dollars, then you will always receive eight dollars. The second equation $5x + 3 = \underline{\quad}$ can model more complex situations. For example, you wish to buy a game that costs \$38 but you only have three dollars. Your uncle will pay you five dollars an hour to work for him. If you've worked five hours, have you earned enough money? If not, how many hours will you have to work?

Algebra and algebraic thinking open up a world of possibilities that arithmetic alone cannot do.